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SIMATS Online Programming Lab

PYTHON PROGRAMMING

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A mathematical analysis system needs to verify whether a given number is prime. A prime number is a number greater than 1 that has no divisors other than 1 and itself.

Write a Python program that:

- Accepts an integer input from the user
- Checks whether the number is prime using appropriate control flow constructs
- Displays a suitable message indicating whether the number is prime or not

Ensure proper use of loops and conditional statements.

Your Code

```
1 n=int(input())
2 prime=True
3 if n<=1:
4     prime=False
5 else:
6     for i in range(2,n):
7         if n%i==0:
8             print=False
9             break
10 if prime:
11     print("Prime")
12 else:
13     print("Not Prime")
```

Input

7

Output

Prime

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A numerical computation module is designed to generate a sequence where each number is the sum of the two preceding numbers. This sequence is known as the Fibonacci series.

Write a Python program that:

- Accepts a number n from the user
- Generates and displays the first n terms of the Fibonacci sequence
- Uses looping constructs to compute the sequence

Ensure the program handles cases where n is less than or equal to zero.

Your Code

```
1 def fib(n):
2     if n<=0:
3         print("Invalid input")
4     else:
5         a=0
6         b=1
7         for i in range(1,n):
8             print(a,end=" ")
9             c=a+b
10            a=b
11            b=c
12            c=a+b
```

Input

5

Output

0 1 1 2 3

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A text processing system receives user input in the form of a string that may contain alphabets, digits, and special characters.

Write a Python program that:

- Accepts a string input from the user
- Counts the number of vowels, consonants, digits, and special characters present in the string
- Displays the count for each category

Use appropriate string operations and control flow statements.

Your Code

```
1 s=input()
2 vowels =0
3 consonants =0
4 digits =0
5 special =0
6 for ch in s:
7     if ch.isalpha():
8         if ch.lower() in "aeiou":
9             vowels +=1
10        else:
11            consonants +=1
12    elif ch.isdigit():
13        digits +=1
14    else:
15        special +=1
```

Input

Hello123#

Output

Vowels =
2
Consonants =
7
Digits =
1

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Analyze the given code and perform the following:

- Identify all errors present in the program
- Rewrite the corrected program
- Briefly justify each correction

```
num = input("Enter number :")
if num % 2 == 0
print("Even")
else
print("Odd")
```

Your Code

```
1 num=int(input("Enter a number: "))
2 if num%2==0:
3     print("Even")
4 else:
5     print("Odd")
```

Input

3

Output

Enter a number: Odd

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A program is intended to display numbers from 1 to 10 using a loop. However, some parts of the code are missing. Complete the program by filling in the blanks so that it correctly prints numbers from 1 to 10.

```
# Program to print numbers from 1 to 10
for i in ____:
    print(____)
```

Also explain how the range() function works in this context.

Your Code

```
1 for i in range(1,11):
2     print(i)
```

Input

stdin input

Output